5. Appendix

A 'real world' problem

In the following continuous model of *epidemic spread*, we consider strictly positive C^∞-functions on a time interval $\mathcal{I} := [\,0\,,\,T\,[\,,\ T \in \mathbb{R}_+ \cup \{+\infty\}$. Remember these functions form a convex cone and a multiplicative group $\mathcal{C} \subset C^\infty(\mathcal{I})$ and $\mathcal{C}_1 = \{\, f \in \mathcal{C} \,|\, f(0) = 1 \,\}$ is a convex subgroup of $\mathcal{C}$. Also, it is easily seen that

$$E \circ J \equiv Id_{\mathcal{C}} \quad \textit{and} \quad J \circ E \restriction \mathcal{C}_1 \equiv Id_{\mathcal{C}_1} \qquad [\text{ cf. (4.1) }]\,, \tag{A.1}$$

where $E : \mathcal{C} \to \mathcal{C}$ and $J : \mathcal{C} \to \mathcal{C}_1$ respectively are the mappings

$$Ef_t = \exp \frac{d}{ds} \log f(s)\,\big|_t \quad \text{and} \quad Jf_t = \exp \int_0^t \log f(s)\, ds\,. \tag{A.2}$$

Remember also that the exponential integral and the exponential mean of f over $[\,a\,,\,b\,] \subset \mathcal{I}$ respectively equal

$$Jf\,|_a^b = \exp \int_a^b \log f(s)\, ds \quad \text{and} \quad \bar{f}_{[a,b]} = \exp \frac{1}{b-a} \int_a^b \log f(s)\, ds\,. \tag{A.3}$$

For simplicity, we normalize the *herd size* to one, hence the *herd infection* is represented by a function $h : \mathcal{I} \to [\,0\,,\,1\,]$ (0: no infections, 1: all members infected). More precisely, $h(t)$ represents the *total infected fraction*, including the acutely sick members and the asymptomatic cases, as well as all who have recovered or who have died (in the SIR model, cf. [SM], this corresponds to the sum $i(t) + r(t)$). The *succeptible* fraction (represented by $s(t)$ in the SIR model) then is obviously $1 - h(t)$. Finally, the portion of *acutely sick* new infections per time unit, i.e. the *incidence*, will be represented by the function $u : \mathcal{I} \to \mathbb{R}_+$.

Usually the *reproduction number* is defined as the number of people that are infected on average by one sick person. By itself, it says thus nothing about the underlying time scale, yet time enters naturally with the *transmission time* T_{tr} which is the average time during that one sick individual infects R susceptible ones. Then obviously $u(\,t + T_{tr}\,) = R \cdot u(\,t\,)$ and we can state the following

Proposition 1 (Reproduction numbers)
With Eu_t as above, the reproduction number, usually defined as $R := \frac{u(\,t+T_{tr}\,)}{u(\,t\,)}$, equals:

$$R = (\,\bar{E}u_{[t,t+T_{tr}]}\,)^{T_{tr}}, \tag{A.4}$$

i.e. $R^{\frac{1}{T_{tr}}}$ is the *exponential mean* of Eu over the transmission interval $[\,t,\,t + T_{tr}\,]$.

Proof : Immediate consequence of Theorem 3 (T3-3).

Remarks

1. Since $Eu : t \mapsto Eu_t$ is continuous, there always exists a time $\tau \in [\,t\,,\,t + T_{tr}\,]$, such that $Eu_\tau = \bar{E}u_{[t,t+T_{tr}]}$, and hence, due to (A.4),
$$R^{\frac{1}{T_{tr}}} = Eu_\tau \tag{A.5}$$
2. Trivially, the transmission time T_{tr} introduces a *natural time scale* in which T_{tr} is the unit of time and hence, for a certain $\tau \in [\,t\,,\,t + T_{tr}\,]$,
$$R = Eu_\tau\ . \tag{A.6}$$

From now on we work in this natural time scale, i.e. with $T_{tr} = 1$. Moreover, we identify $R = R_t$ with $Eu_t = \exp \frac{d}{dt} \log u(t)$, which boils down to refining the definition of R by replacing the exponential mean of Eu over a transmission cycle by its *actual* value at time t. This exactly is summarized in the following

Proposition 2 . In the natural time scale (in which $T_{tr} = 1$), the reproduction number is the exponential derivative of the incidence.

Every mathematical model makes simplifications. An essential assumption of ours is an *homogeneous environment*. This means, to put it simply, that the pathogene finds the same conditions for spreading throughout the herd. In particular, singular spreading events or new infections from outside are neglected. As far as these conditions are met, the essence of the disease spread is captured in the following equations.

The infected fraction h at time t is

$$h(t) \;=\; h(0) \;+\; P \int_0^t u(\tau) d\tau, \qquad \text{with} \qquad P \;=\; \frac{A}{100 - B}\;, \tag{A.7}$$

P being the factor that takes the percentage A of *steril-immunized* and B of *asymptomatic* ('silent') cases into account (so if all infected survivors become non-infectiously immune, then obviously $A = 100$). Since $u(t)$ is strictly positive, $h(t)$ is strictly increasing and so the susceptible fraction $s(t) \;=\; 1 - h(t)$ is strictly decreasing.
Under the assumed homogeneous conditions, the exponential derivative of the incidence obviously decreases linearly with the susceptibility, i.e.

$$\begin{aligned} Eu_t \;&=\; s(t)\; Eu_0 \;=\; [\,1 \;-\; h(t)\,]\, R_0 \\ &=\; [\,1 - h(0) - P \int_0^t u(\tau) d\tau\,]\, R_0 \end{aligned} \tag{A.8}$$

and so the incidence becomes, due to (A.1) and (EI),

$$\begin{aligned} u(t) \;&=\; u(0)\; J\, Eu_t \;=\; u(0)\, \exp \int_0^t \log Eu_\tau \, d\tau \\ &(\;\wr\; J \circ Eu = u/u(0)\;) \\ &=\; u(0)\; \exp \int_0^t \log \{\, [\,1 - h(0) - P \int_0^\tau u(\theta)\, d\theta\,]\, R_0 \,\}\; d\tau\,, \end{aligned} \tag{A.9}$$

where $u(0)$, $h(0)$ and R_0 represent respectively the initial incidence, the initial herd infection and the reproduction number at time $t = 0$.

Equation (A.9) describes the disease spread in a static homogeneous environment. The exorbitant dynamics that an epidemic can develop at higher reproduction numbers is also seen in the derivative of $u(t)$ obtained from (A.9)

$$\frac{du}{dt} \;=\; u(t)\, \log \{\, [\,1 - h(0) - P \int_0^t u(\tau)\, d\tau\,]\, R_0 \,\}\,. \tag{A.10}$$

The graphics below display in linear and logarithmic scale numerical solutions of equation (A.9) for basic reproduction numbers varied from $R_0 = 0.5$ to $R_0 = 3.0$ and initial conditions $h(0) = 0$ and $u(0) = 10^{-5}$ (in common parlance: 'incidence one', i.e. one sick person in a population of 100.000). We also assume $A = 100\%$, that is every surviving infected person ends up non-infectiously immune, and $B = 20\%$ the percentage of asymptomatic cases.

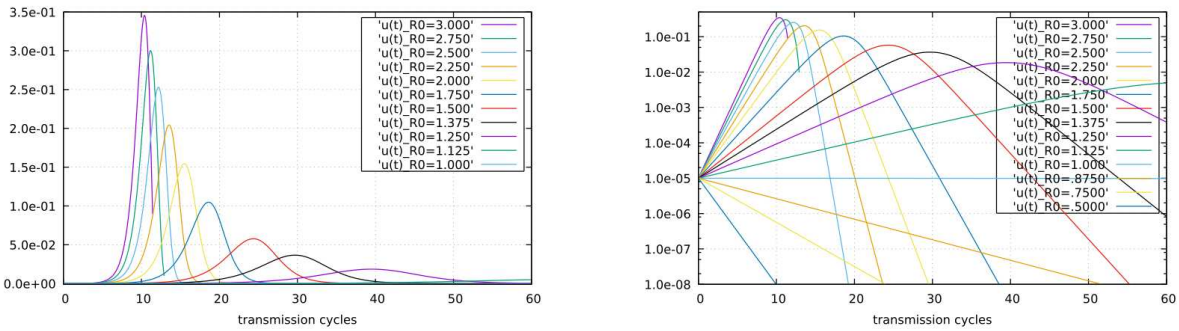

Fig. 1. *The incidence*

The graphics give an idea of what is happening if an epidemic with higher reproduction number is let loose, i.e. if it is not contained by suitable measures. For $R_0 \gtrsim 2.7$, a considerable part of the population would get simultaneously sick at the peak.

With initially $R_0 \gtrsim 2.7$, everybody would eventually have gone through the infection, be it alive or dead. The entire population would end up immune (the cuts in the two upper curves indicate when 100% herd immunity is attained). Assuming a lethality of 0.2% (estimated for Covid-19, e.g.) in a population of 82 million as in Germany more than 160,000 people would die. One can only hope for any effective vaccination, for otherwise the reproduction number has to be kept low with painful containment measures.

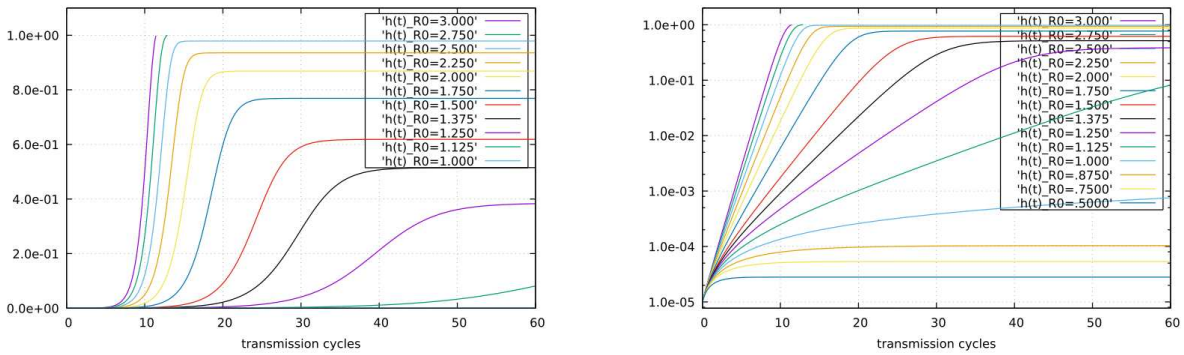

Fig. 2. *The herd infection*

6. Conclusions and outlook

The exponential derivative of a differentiable path $c : \mathcal{I} \to G$ in a Lie group has been studied and it was demonstrated how it acts *within* the group through right or left multiplication and in this way provides an *intrinsic* description of the infinitesimal change, in contrast to the classical differential that acts in *tangent* space.

Basic properties have been outlined and calculation rules derived. The elementary case of the multiplicative group $G = \mathbb{R}_+$ together with the smooth strictly positive functions $f : \mathcal{I} \to \mathbb{R}_+$ was treated in more detail, and applied to a numerical model of epidemic spread.

We are particularly interested in novel methods and applications in dynamic systems with many interacting components. The general framework in finite or infinite dimensional Lie groups may be of interest here. However, this must be reserved for further investigations.

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STEFFEN HEIN; DE-83043 Bad Aibling, Bavaria / Germany, `https://www.sfenx.de`
E-mail address: contact@sfenx.de